Moving to MongoDB Atlas

The Pets application with a remote database

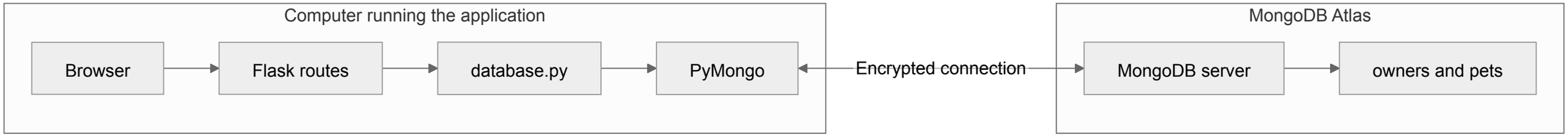
Gregory S. DeLozier, PhD

The Database Moves

Mongita	MongoDB Atlas
Embedded Python database	Hosted MongoDB server
Local disk directory	Remote cluster
Mongita client	PyMongo driver
Local file access	Network connection and login

The app still calls `database.py`.

Where the Code Runs



Atlas Names

Name	Meaning in this exercise
Atlas account	Login to the management website
Project	`Advanced-Database`
Cluster	`course-pets`
Database	`pets`
Database user	`user`

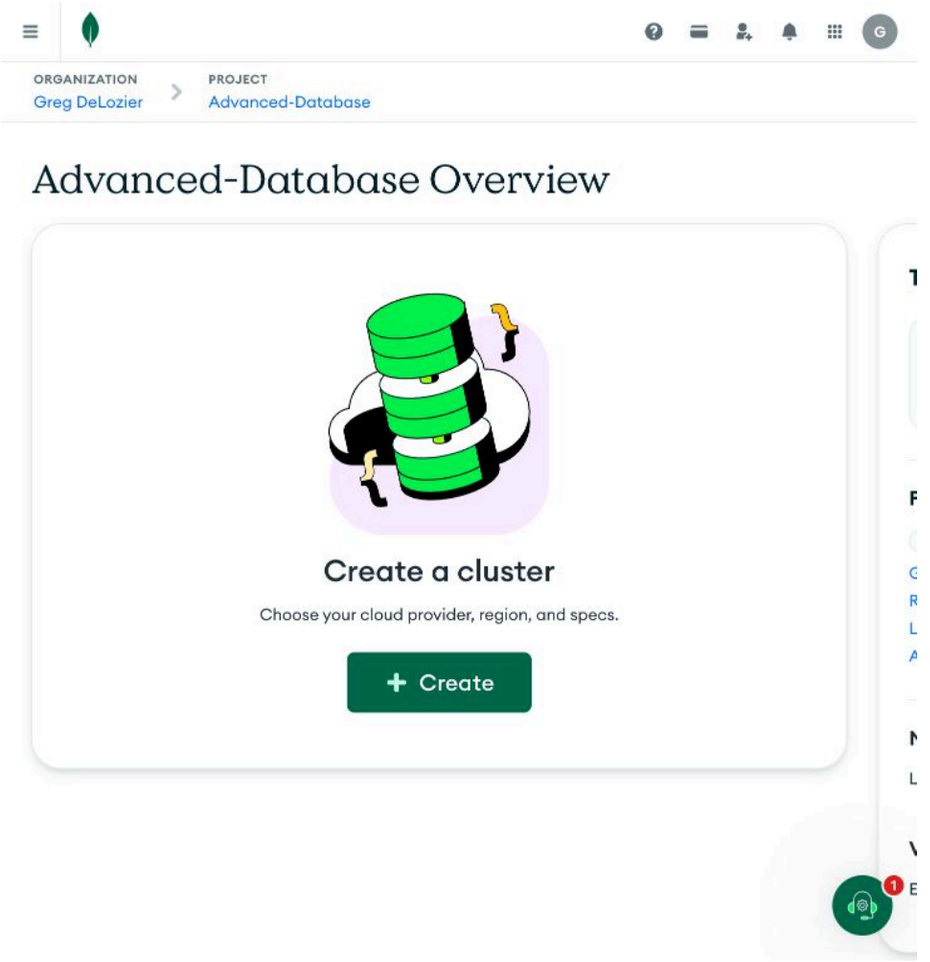
Log In and Create a Project

Open **https://cloud.mongodb.com/**.

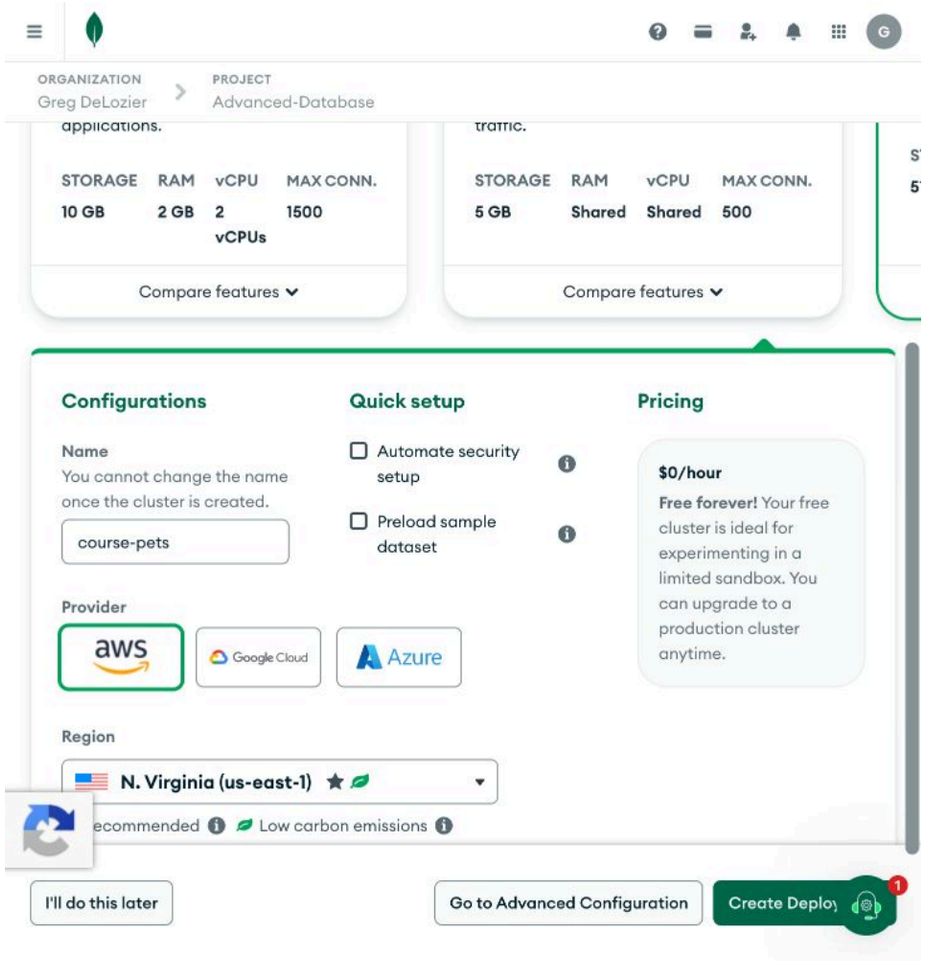
Sign in or create an account.

Choose **New Project** and name it **Advanced-Database**.

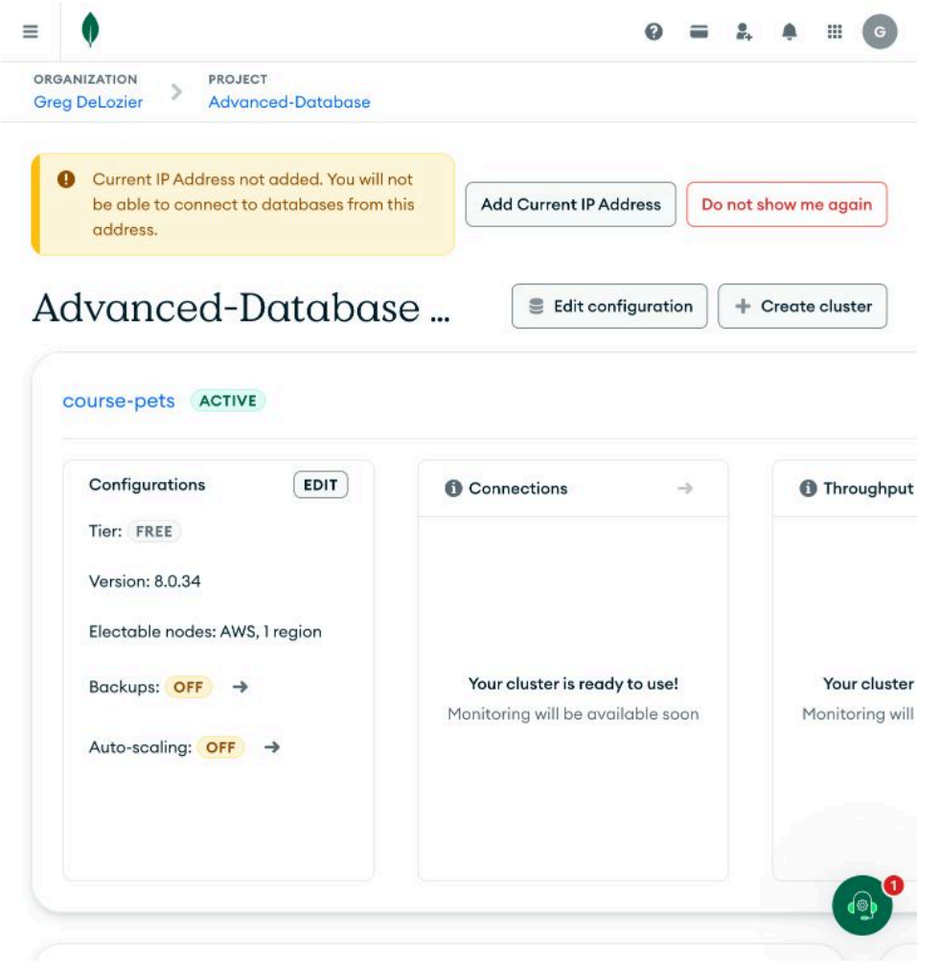
Project Ready for a Cluster



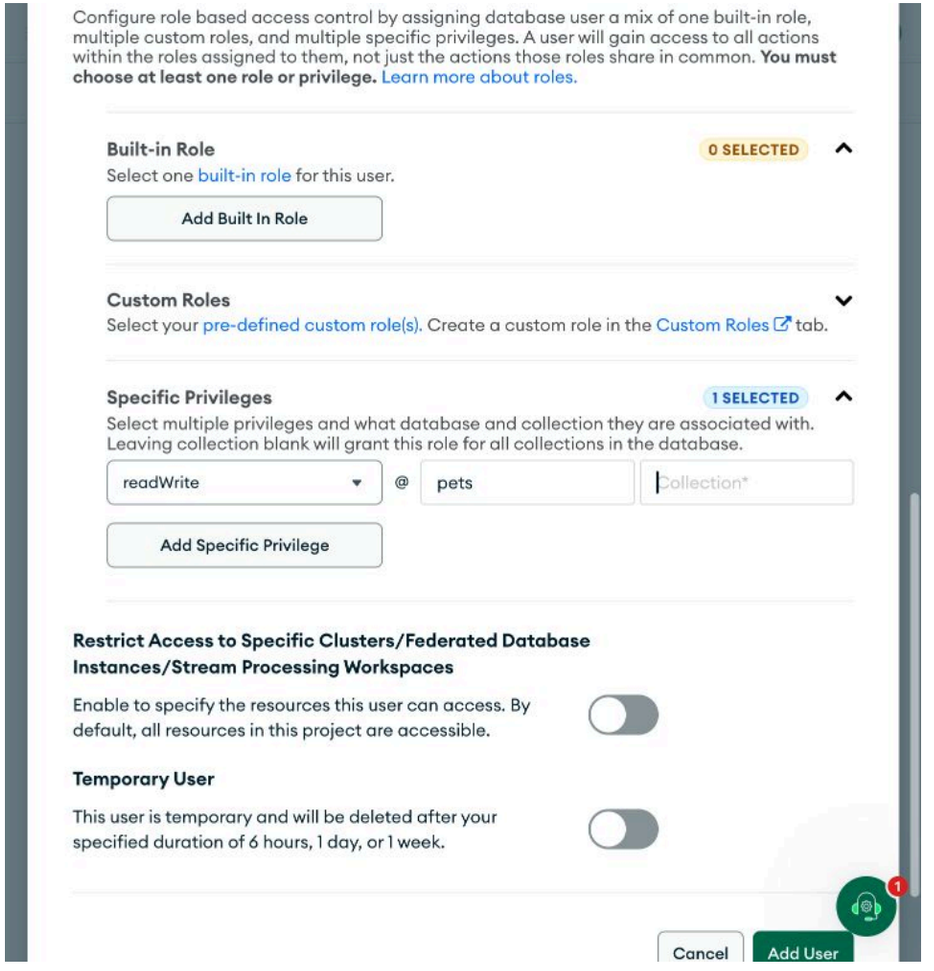
Select the Free Tier



Wait for the Cluster



Database User Privileges

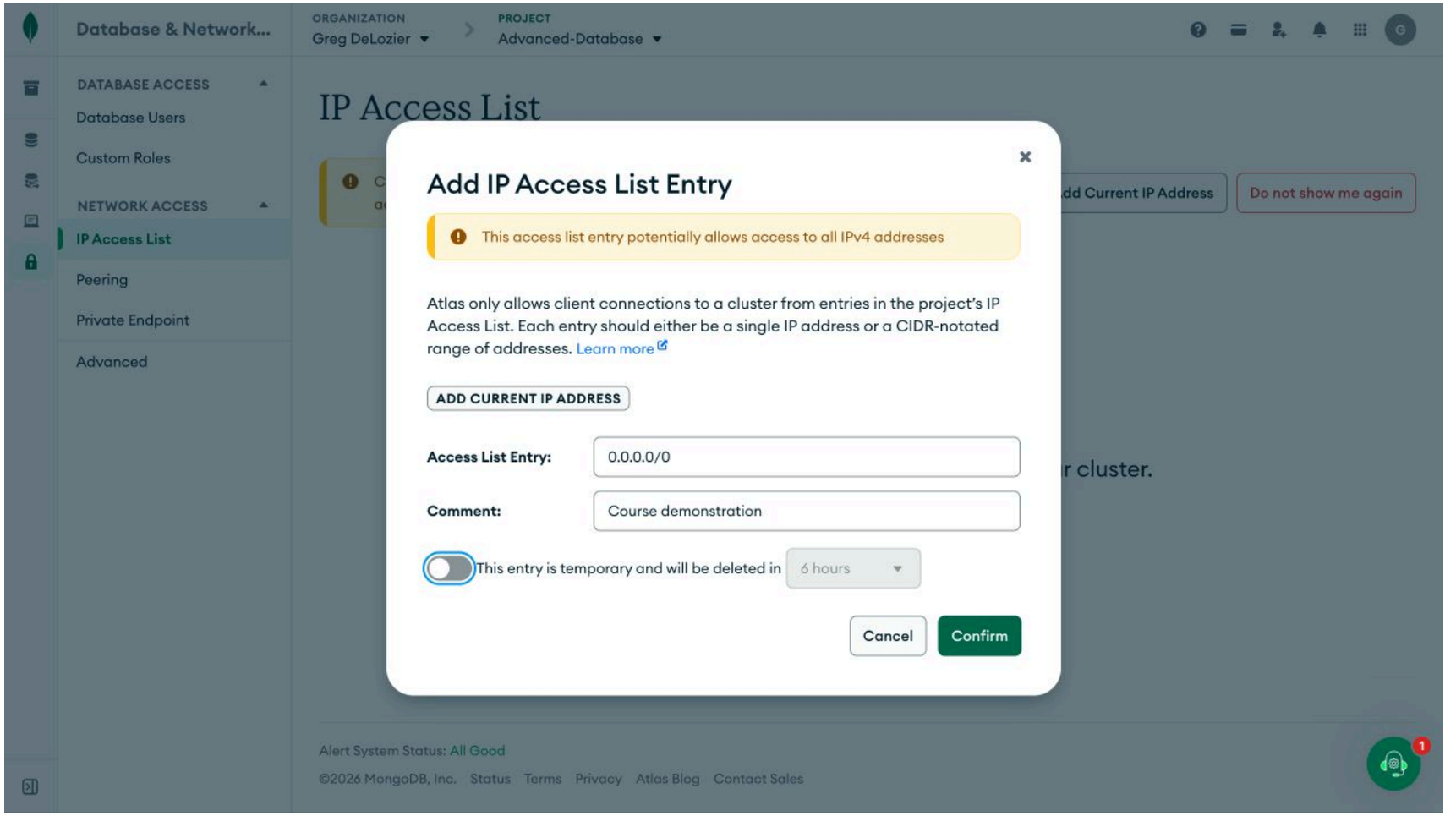


The Classroom Login

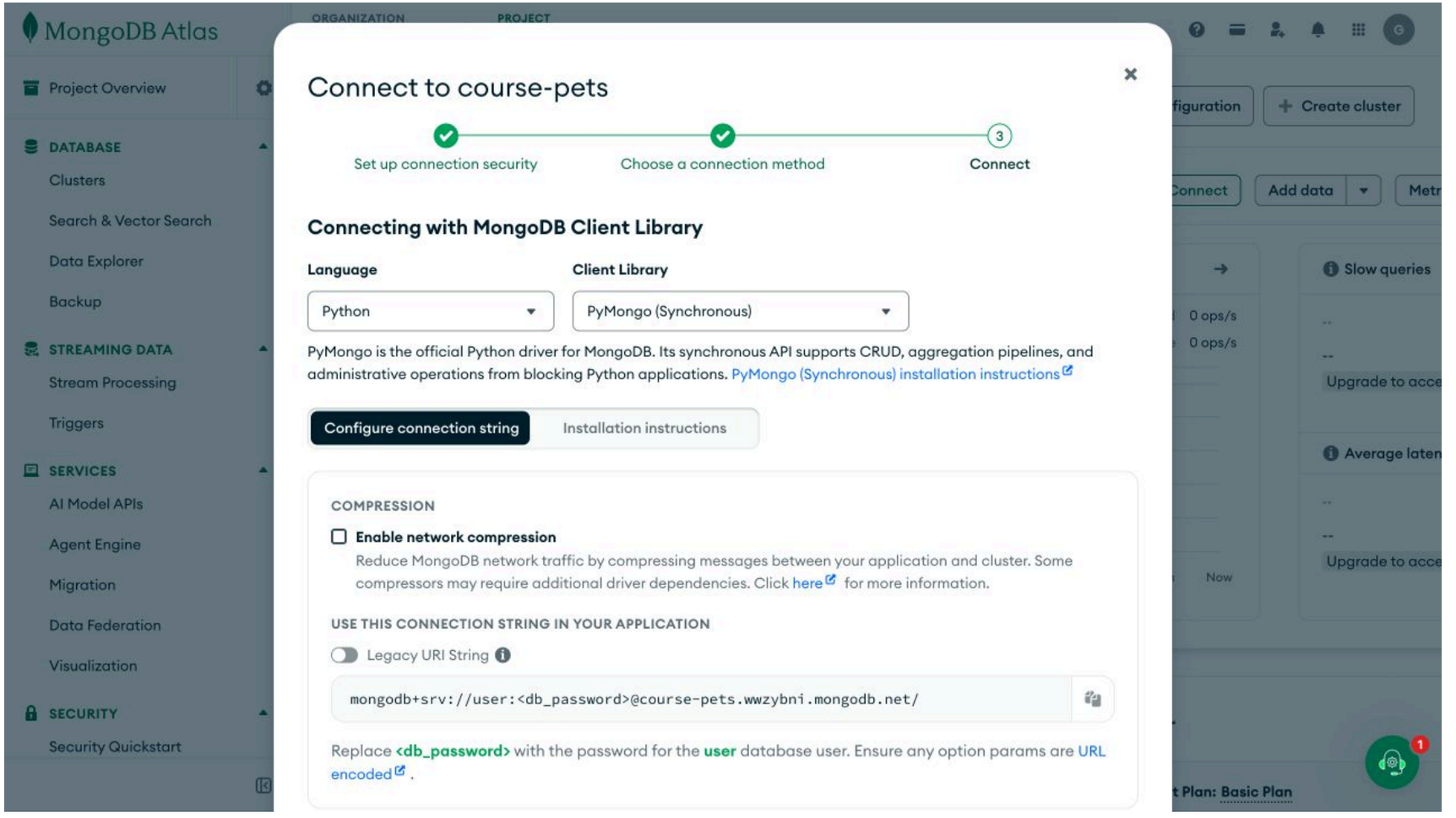
Setting	Value
Username	`user`
Password	`alphabeta123`
Role	`readWrite` on `pets`

Temporary, shared exercise credentials.

Allow Classroom Connections



Select the Python Driver



The Classroom Connection String

```
mongodb+srv://user:alphabeta123@course-pets.wwzybni.mongodb.net/
```

This walkthrough deployment has been deleted.

Use your own cluster hostname and credentials in the commands.

Prepare Python

```
cd chapter-13-mongo-atlas/code
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

Python **3.10 or newer**. Dependencies: Flask and PyMongo.

Set the Connection Environment

Bash:

```
export MONGODB_URI='mongodb+srv://user:alphabeta123@course-
pets.wwzybni.mongodb.net/'
```

PowerShell:

```
$env:MONGODB_URI = 'mongodb+srv://user:alphabeta123@course-
pets.wwzybni.mongodb.net/'
```

Substitute your own connection string. Use **the same terminal**.

Check the Connection

```
python access-atlas-demo.py
```

```
Connected to MongoDB; pets database is readable.
Owners: 0
Pets: 0
```

Existing records produce different counts.

Inside the Connection Code

```
uri = os.environ.get("MONGODB_URI")
client = MongoClient(
    uri, server_api=ServerApi("1"),
    serverSelectionTimeoutMS=10000
)
client.admin.command("ping")
```

Connect at startup and check the result.

Select the Collections

```
db = client[database_name]
owners_collection = db.owners
pets_collection = db.pets
```

`setup_database("pets")` opens the application database.

IDs still cross the layer as strings.

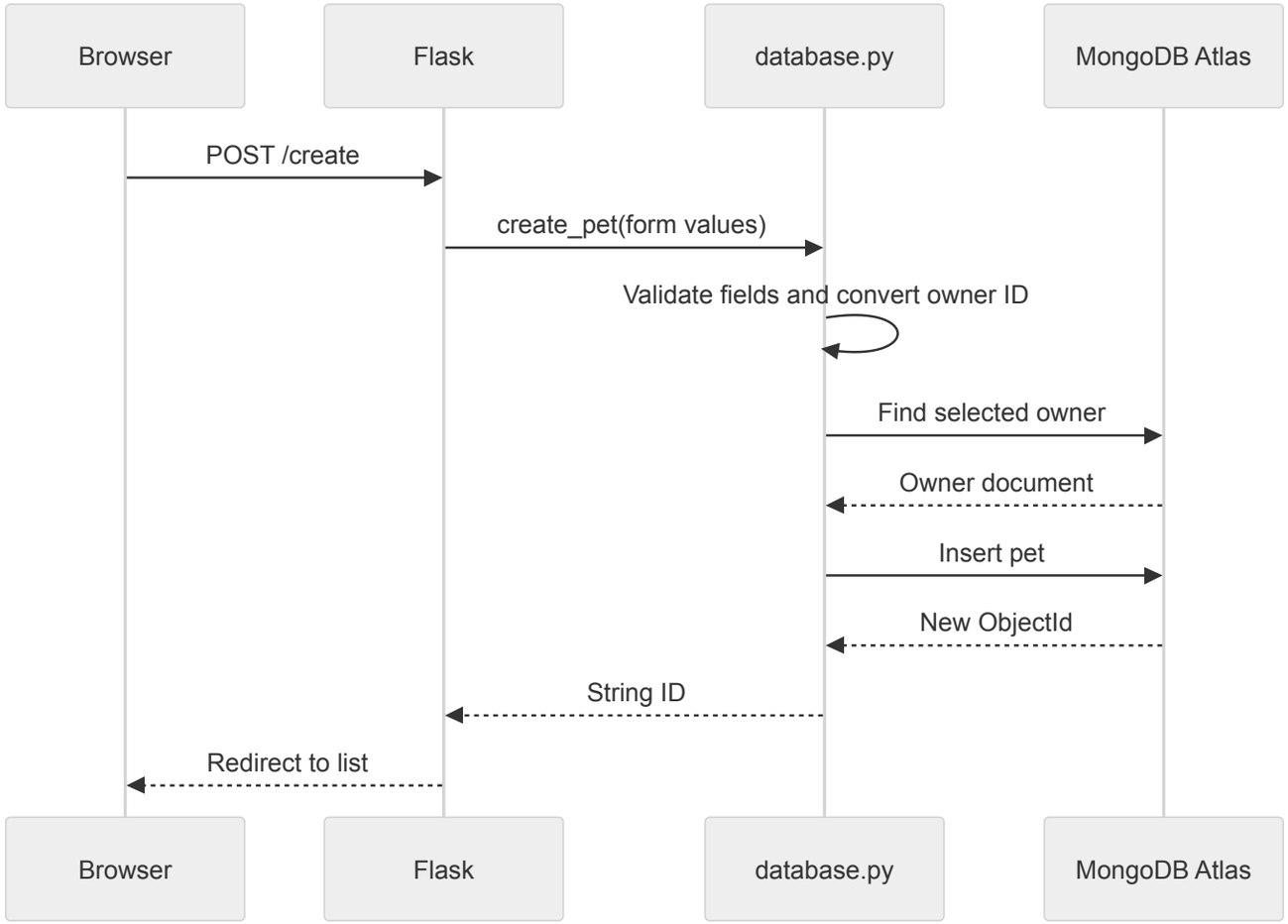
Run the Pets Application

```
flask --app app run --debug
```

Open the local address printed by Flask.

Create **Alex**, then **Casey**, age **3**, owned by Alex.

A Save Reaches Atlas



Inspect the Stored Documents

In Atlas, open **Data Explorer**.

Select **course-pets**, then **pets**.

Inspect the **owners** and **pets** collections.

Casey's `owner_id` matches Alex's `_id`.

Integrity Rules Stay in Our Layer

Rule	Code responsibility
Nonblank name and type	Validate before saving
Non-negative age	Reject invalid values
Pet must have an owner	Look up the owner
Owner with pets cannot be deleted	Check dependent pets

Atlas does not infer these application rules.

MongoDB Supports ACID Transactions

Atomicity, Consistency, Isolation, Durability

A transaction can group changes across documents and collections.

Our Pets app currently sends **separate operations**.

Eventual Consistency

The primary accepts writes; secondaries replicate them.

A secondary may temporarily show an older value.

When writes stop and replication proceeds, copies converge.

Verify the Same Behavior

Create **Blair**. Reassign Casey to Blair, age **4**.

Submit age **-5**: the saved age remains **4**.

Try deleting Blair: the app rejects it while Casey refers to Blair.

A New Connection Does Not Copy Data

The new connection opens the **Atlas database**.

Existing Mongita files stay on the local computer.

Moving old records needs a separate data-copy program.

Check Failures at the Right Place

Result	First check
Missing `MONGODB_URI`	Current terminal environment
Authentication failed	Database username and password
Connection timeout	Network access list and hostname
Invalid age or missing owner	Form values and app rules

Clean Up After the Exercise

Delete the temporary database user.

Remove the `0.0.0.0/0` network entry.

Or terminate the exercise cluster when its data is no longer needed.

Do not leave shared classroom access active in service.